



DATA SCIENCE FIELD NOTES / 03C

FEATURE ENGINEERING DOCUMENTATION

Twelve interpretable features, controlled redundancy and leakage-safe preprocessing

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	Loan Approval Analysis and Feature Engineering
PHASE	Week 3 Completed analytical preparation

Educational portfolio report. Historical approval patterns do not constitute a production lending model or lending policy.

Purpose and scope

Feature engineering improves the interpretability and modelling usefulness of the upstream cleaned dataset. The work introduces 12 representations covering dependents, household structure, income segmentation, coapplicant contribution, repayment burden, credit interpretation and financial transformations.

Design principle: a useful engineered feature needs a clear business interpretation, valid construction, defensible modelling benefit and documented limitation.

Project measure	Verified result
Upstream cleaned observations	614
Upstream cleaned columns	14
Engineered features	12
Final analytical columns	26
Selected source modelling features	12
Final encoded predictors	13

Data and unit considerations

The upstream `loan_income_ratio` divides a loan amount recorded in thousands by monthly income in currency units, so the units are not directly comparable. The improved payment-income proxy first converts the recorded amount into currency units and then divides it by the repayment term.

Feature catalogue - Household and income

Feature	Construction	Interpretation
dependents_numeric	Map 3+ conservatively to 3	Machine-readable dependent count
family_size	1 + married indicator + dependents	Approximate household responsibility
family_size_group	Small / Medium / Large	Readable household segment
income_band	Low / Middle / High / Very High	Readable income segment
has_coapplicant_income	1 if coapplicant income > 0	Single- vs dual-income signal
coapplicant_income_share	Coapplicant income / total income	Dependence on shared income

Interpretation notes

The 3+ dependents category is mapped to 3 because the exact value is unavailable, so family size remains an approximation. Fixed income-band thresholds require review before use in another market or currency context. Zero coapplicant income is a valid observation, not a missing value.

Selection decision

- Retain `dependents_numeric`, `has_coapplicant_income` and `coapplicant_income_share` for modelling.
- Keep `family_size`, `family_size_group` and `income_band` for analytical reporting.
- Avoid combining `family_size` with every exact source component when unnecessary.

Feature catalogue - Credit, duration and affordability

Feature	Construction	Interpretation
term_years	Loan term in months / 12	Readable duration
estimated_monthly_principal	Loan amount x 1,000 / term	Approximate principal instalment
payment_income_ratio	Estimated principal / total income	Repayment-burden proxy
credit_risk_category	Readable label from credit history	Stakeholder-facing category
log_total_income	Natural log of 1 + total income	Reduced income skewness
log_loan_amount	Natural log of 1 + loan amount	Reduced amount skewness

Affordability proxy example

For the first application, a recorded amount of 128 represents approximately 128,000 currency units. Dividing by a 360-month term gives estimated principal of 355.56 per month. Dividing by total income of 5,849 produces a payment-income proxy of 6.08%.

The proxy excludes interest, insurance, existing debt and other obligations. It is suitable as an analytical feature, not as a complete underwriting decision.

Redundancy controls

- Retain credit_history for modelling and credit_risk_category for reporting.
- Retain term_years instead of duplicating the original month-based term.
- Use log_total_income and log_loan_amount instead of their raw equivalents in the model inputs.
- Represent estimated monthly principal within payment_income_ratio rather than retaining both.

Final selected modelling inputs

Feature family	Selected inputs
Applicant context	married; dependents_numeric; education; self_employed
Location and credit	property_area; credit_history
Financial transformations	log_total_income; log_loan_amount; term_years
Household and affordability	has_coapplicant_income; coapplicant_income_share; payment_income_ratio

Responsible modelling decision

Gender is excluded from predictive inputs because it can act as a protected characteristic and reproduce historical bias. It remains available in the analytical dataset for subgroup fairness auditing.

Leakage-safe preprocessing

- Split 614 applications into 491 training and 123 testing rows using stratified sampling.
- Fit one-hot encoding on training categories only and ignore unseen test categories.
- Fit standard scaling on training values of continuous engineered features only.
- Pass binary and ordinal variables through without unnecessary scaling.
- Publish cleaned, combined ML-ready, training and testing datasets with a data dictionary.

Conclusion

The final feature set combines statistical evidence with interpretable financial relationships, controlled redundancy, fairness awareness and reproducible train/test preprocessing. It prepares the project for careful model comparison without claiming production readiness.